RSpec Predefined Matchers

Before you run the tests, let's take a quick look at RSpec's basic matchers. RSpec predefines a number of matchers. What follows is a list of the most useful ones; you can find a full list online.[2]

```
expect(array).to all(matcher)
expect(actual).to be > expected     # (also works with <, >=, <=, and ==)
expect(actual).to be_a(type)
expect(actual).to be_truthy
expect(actual).to be_falsy
expect(actual).to be_nil
expect(actual).to be_between(min, max)
expect(actual).to be_within(delta).of(expected)
expect { block }.to change(receiver, message, &block)
expect(actual).to contain_exactly(expected)
expect(range).to cover(actual_value)
expect(actual).to eq(expected)
expect(actual).to exist
expect(actual).to have_attributes(key/value pairs)
expect(actual).to include(*expected)
expect(actual).to match(regex)
expect { block }.to output(value).to_stdout   # also to_stderr
expect { block }.to raise_error(exception)
expect(actual).to satisfy { block }
```

Most of these mean what they appear to say. Some elaborations:

- The all matcher takes a different matcher as an argument and passes if all elements of the array pass that internal matcher, as in expect([1, 2, 3]).to all(be_truthy).
- The change matcher takes a block argument that passes if the value of receiver.message changes when the block is evaluated.
- The contain_exactly matcher is true if the expected array and the actual array contain the same elements, regardless of order.
- The satisfy matcher passes if the block evaluates to true.
- The matchers that take block arguments, output and raise_error, are for specifying a side effect of the block's execution—that it raises an error or that it changes a different value—rather than the state of a particular object.

2. https://relishapp.com/rspec/rspec-expectations/v/3-7/docs/built-in-matchers